

LC79H (AL)

Hardware Design

GNSS Module Series

Version: 1.2

Date: 2023-04-26

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2023. All rights reserved.

Safety Information

The following safety precautions must be observed during all phases of operation, such as usage, service, or repair of any terminal or mobile incorporating the module. Manufacturers of the terminal should notify users and operating personnel of the following safety information by incorporating these guidelines into all product manuals. Otherwise, Quectel assumes no liability for customers' failure to comply with these precautions.



Ensure that the product may be used in the country and the required environment, as well as that it conforms to the local safety and environmental regulations.



Keep away from explosive and flammable materials. The use of electronic products in extreme power supply conditions and locations with potentially explosive atmospheres may cause fire and explosion accidents.



The product must be powered by a stable voltage source, and the wiring shall conform to security precautions and fire prevention regulations.



Proper ESD handling procedures must be followed throughout the mounting, handling and operation of any devices and equipment that incorporate the module to avoid ESD damages.

About the Document

Document Information	
Title	LC79H (AL) Hardware Design
Subtitle	GNSS Module Series
Document Type	Hardware Design
Document Status	Released

Revision History

Version	Date	Description
-	2021-05-31	Creation of the document
1.0	2022-03-30	First official release
1.1	2023-03-02	- Deleted the information of LC79H (BA). - Added the voltage of V_BCKP and the number of concurrent GNSS, and updated the weight of LC79H (AL) (Table 2). - Added the power data, TTFF with EASY and EPO, and updated the accuracy of 1PPS signal and the velocity accuracy (Table 3). - Added a notch circuit in the block diagram (Figure 1). - Added a software command to output the geofence status (Chapter 1.9). - Added the DC Characteristics of all pins, and updated supported messages for I2C interface (Table 6). - Updated the supported messages and baud rate for UART (Table 6 and Chapter 4.1.1.1). - Added the reference power supply circuit with 3.7 V lithium battery and a note that it is not recommended to use a non-rechargeable battery to power V_BCKP in Continuous mode (Chapter 3.2.2). - Added the optional SAW filter circuit to active and passive antenna reference designs, as well as the corresponding description (Chapter 5.2).

Version	Date	Description
		10. Updated the maximum input power at RF_IN (Table 10). 11. Added the high-level input voltage range of RESET_N and maximum output current of VDD_RF (Table 11). 12. Added the module mounting direction (Chapter 8.1.3). 13. Added the sizes of pizza box and carton (Chapter 8.1.4). 14. Updated the recommended ramp-to-soak, ramp-up and cool-down slopes (Figure 26 and Table 15).
1.2	2023-04-26	Added the maximum update rate for NMEA message and supported update rate for RTCM message (Table 3).

Contents

Safety Information.....	3
About the Document.....	4
Contents.....	6
Table Index.....	8
Figure Index.....	9
1 Product Description.....	10
1.1. Overview.....	10
1.1.1. Special Mark.....	10
1.2. Features	11
1.3. Performance.....	13
1.4. Block Diagram	14
1.5. GNSS Constellations	14
1.6. Augmentation System	15
1.6.1. SBAS.....	15
1.7. AGNSS.....	15
1.7.1. EASY.....	15
1.7.2. EPO.....	16
1.8. Multi-Tone AIC	16
1.9. Geofencing.....	16
1.10. Firmware Upgrade.....	17
2 Pin Assignment.....	18
3 Power Management	22
3.1. Power Unit.....	22
3.2. Power Supply	23
3.2.1. VCC.....	23
3.2.2. V_BCKP	23
3.3. Power Modes	25
3.3.1. Feature Comparison	25
3.3.2. Continuous Mode	25
3.3.3. Backup Mode	25
3.4. Power-up Sequence.....	26
3.5. Power-down Sequence	27
4 Application Interfaces	28
4.1. I/O Pins.....	28
4.1.1. Communication Interfaces	28
4.1.1.1. UART Interface.....	28
4.1.1.2. I2C Interface	29
4.1.2. 3D_FIX.....	30
4.1.3. GEOFENCE	30

4.1.4.	JAM_IND	30
4.1.5.	1PPS	30
4.2.	System Pins	30
4.2.1.	WAKEUP	30
4.2.2.	RESET_N	30
5	Design	32
5.1.	Antenna Selection	32
5.1.1.	Antenna Specifications	32
5.1.2.	Antenna Selection Guide	33
5.2.	Antenna Reference Design	33
5.2.1.	Active Antenna Reference Design	33
5.2.2.	Passive Antenna Reference Design	35
5.3.	Coexistence with Cellular Systems	35
5.3.1.	In-band Interference	35
5.3.2.	Out-of-band Interference	37
5.3.3.	Ensuring Interference Immunity	37
5.4.	Recommended Footprint	39
6	Electrical Specification	40
6.1.	Absolute Maximum Ratings	40
6.2.	Recommended Operating Conditions	40
6.3.	Supply Current Requirement	41
6.4.	ESD Protection	42
7	Mechanical Dimensions	43
7.1.	Top, Side and Bottom View Dimensions	43
7.2.	Top and Bottom Views	44
8	Product Handling	45
8.1.	Packaging	45
8.1.1.	Carrier Tape	45
8.1.2.	Plastic Reel	46
8.1.3.	Mounting Direction	46
8.1.4.	Packaging Process	47
8.2.	Storage	48
8.3.	Manufacturing and Soldering	49
9	Labelling Information	51
10	Appendix References	52

Table Index

Table 1: Special Mark.....	10
Table 2: Product Features.....	11
Table 3: Product Performance	13
Table 4: GNSS Constellations and Frequency Bands.....	15
Table 5: I/O Parameter Definition	19
Table 6: Pin Description	19
Table 7: Feature Comparison in Different Power Modes	25
Table 8: Recommended Antenna Specifications.....	32
Table 9: Intermodulation Distortion (IMD) Products.....	36
Table 10: Absolute Maximum Ratings	40
Table 11: Recommended Operating Conditions.....	41
Table 12: Supply Current	42
Table 13: Carrier Tape Dimension Table (Unit: mm).....	45
Table 14: Plastic Reel Dimension Table (Unit: mm).....	46
Table 15: Recommended Thermal Profile Parameters	50
Table 16: Related Documents.....	52
Table 17: Terms and Abbreviations	52

Figure Index

Figure 1: Block Diagram.....	14
Figure 2: Pin Assignment.....	18
Figure 3: Internal Power Supply Block Diagram.....	22
Figure 4: VCC Input Reference Circuit.....	23
Figure 5: Backup Domain Reference Circuit.....	24
Figure 6: Reference Power Supply Circuit with 3.7 V Lithium Battery.....	24
Figure 7: Enter/Exit Backup Mode Sequence.....	26
Figure 8: Power-up Sequence.....	27
Figure 9: Power-down and Power-on Restart Sequence.....	27
Figure 10: UART Interface Reference Design.....	28
Figure 11: I2C Interface Reference Design.....	29
Figure 12: Reference OC Circuit for Module Reset.....	31
Figure 13: Reset Sequence.....	31
Figure 14: Active Antenna Reference Design.....	34
Figure 15: Passive Antenna Reference Design.....	35
Figure 16: In-band Interference on GPS L1.....	36
Figure 17: Out-of-band Interference on GPS L1.....	37
Figure 18: Interference Source and Its Path.....	38
Figure 19: Recommended Footprint.....	39
Figure 20: Top, Side and Bottom View Dimensions.....	43
Figure 21: Top and Bottom Views.....	44
Figure 22: Carrier Tape Dimension Drawing.....	45
Figure 23: Plastic Reel Dimension Drawing.....	46
Figure 24: Mounting Direction.....	46
Figure 25: Packaging Process.....	47
Figure 26: Recommended Reflow Soldering Thermal Profile.....	49
Figure 27: Labelling Information.....	51

1 Product Description

1.1. Overview

Quectel LC79H (AL) module is a dual-band, multi constellation receiver with enhanced performance. It can track satellite constellations, such as GPS, GLONASS, Galileo, BDS and QZSS, and also support SBAS (including WAAS, EGNOS, MSAS and GAGAN) and AGNSS functions. The module is powered by 1.8 V.

Key features:

- The dual-band, multi-constellation GNSS module features a high-performance, high reliability positioning engine facilitating fast and precise GNSS positioning.
- Supported serial communication interfaces: UART and I2C.
- Embedded flash memory provides the capacity for storing not only user-specific configurations, but also future firmware upgrades.

Quectel LC79H (AL) is a SMD type module with a compact form factor of 10.1 mm × 9.7 mm × 2.4 mm. It can be embedded in your applications through the 28 pins, including 18 LCC pins and 10 LGA pins.

The module is fully compliant with the EU RoHS Directive.

1.1.1. Special Mark

Table 1: Special Mark

Mark	Definition
●	The symbol indicates that a function or technology is supported by the module(s).

1.2. Features

Table 2: Product Features

Features		LC79H (AL)
Grade	Industrial	●
	Automotive	-
Category	Standard Precision GNSS	●
	High Precision GNSS	-
	DR	-
	RTK	-
	Timing	-
VCC Voltage	1.75–1.98 V, Typ. 1.8 V	●
V_BCKP Voltage	1.62–1.98 V, Typ. 1.8 V	●
I/O Voltage	Same as VCC	●
Communication Interfaces	UART	●
	SPI	-
	I2C	●
	CAN	-
	USB	-
Integrated Features	Additional LNA	●
	Additional Filter	●
	RTC Crystal	●
	TCXO Oscillator	●
	6-axis IMU	-
Constellations and Frequency Bands	Number of concurrent GNSS	4 + QZSS
	GPS L1 C/A	●

Features		LC79H (AL)
	L5	●
	L2C	-
	GLONASS	L1
		L2
	Galileo	E1
		E5a
		E5b
	BDS	B1I
		B1C
		B2a
		B2I
	QZSS	L1 C/A
		L5
		L2C
	NavIC	L5
	SBAS	L1
Temperature Range		Operating temperature range: -40 °C to +85 °C Storage temperature range: -40 °C to +90 °C
Physical Characteristics		Size: (10.1 ±0.15) mm × (9.7 ±0.15) mm × (2.4 ±0.20) mm Weight: Approx. 0.5 g

NOTE

For more information about GNSS constellation configurations, see [document \[1\] protocol specification](#).

1.3. Performance

Table 3: Product Performance

Parameter	Specification	LC79H (AL)
Power Consumption ¹ (GPS + GLONASS + Galileo + BDS + QZSS)	Acquisition	33 mA (59.4 mW)
	Tracking	33 mA (59.4 mW)
	Backup mode	20 μ A (36 μ W)
Sensitivity	Acquisition	-148 dBm
	Reacquisition	-159 dBm
	Tracking	-166 dBm
TTFF ¹ (without AGNSS)	Cold Start	26 s
	Warm Start	18 s
	Hot Start	1 s
TTFF ² (with EASY)	Cold Start	6 s
	Warm Start	2 s
	Hot Start	1 s
TTFF ² (with EPO)	Cold Start	5 s
Horizontal Position Accuracy ³		1.0 m
Update Rate	NMEA	1 Hz (Max. 10 Hz)
	RTCM	1 Hz
Accuracy of 1PPS Signal ¹	RMS	10 ns
	3 σ	15 ns
Velocity Accuracy ¹		Without Aid: 0.03 m/s
Acceleration Accuracy ¹		Without Aid: 0.1 m/s ²
Dynamic Performance ¹		Maximum Altitude: 10000 m Maximum Velocity: 500 m/s Acceleration: 4g

¹ Room temperature, all satellites at -130 dBm.

² Open sky, active high-precision GNSS antenna.

³ CEP, 50 %, 24 hours static, -130 dBm, more than 6 SVs.

1.4. Block Diagram

A block diagram of the module is presented below, which includes a front-end section with a notch circuit, an additional LNA and a diplexer, a TCXO, an XTAL, and a GNSS IC with a PMU. The diplexer integrates two band-pass filters, which can improve the out-of-band rejection.

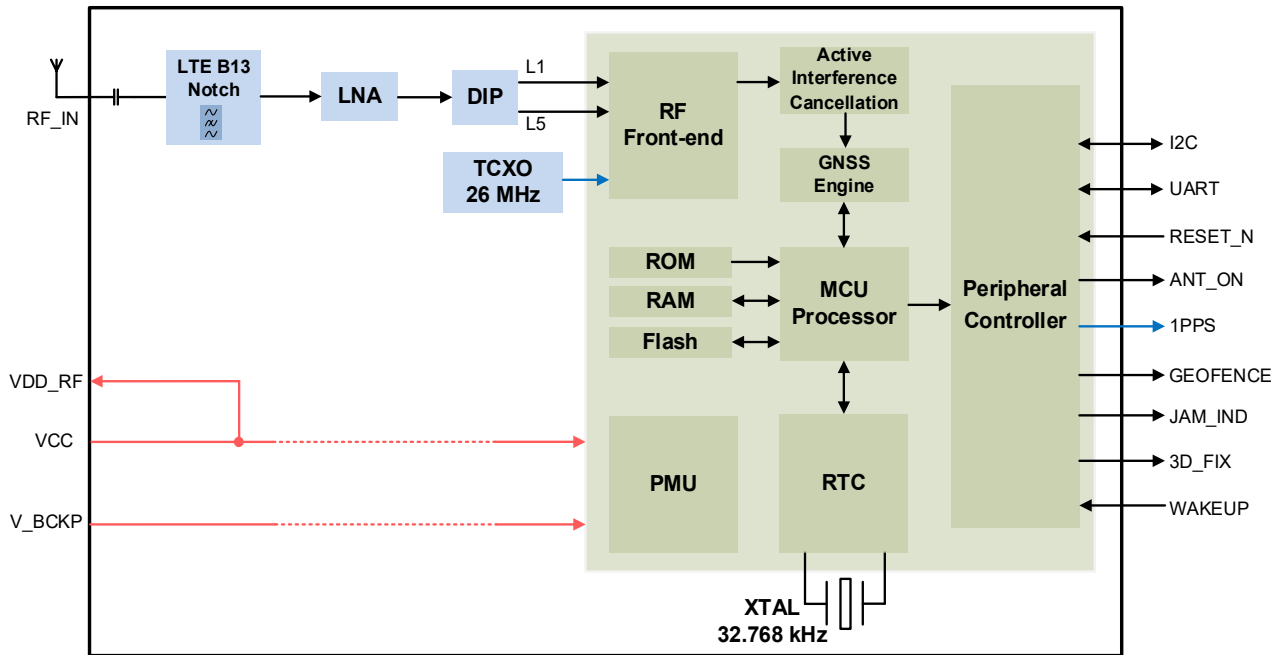


Figure 1: Block Diagram

1.5. GNSS Constellations

The module is a dual-band concurrent GNSS receiver that can receive and track multiple GNSS constellations. It can concurrently track the following GNSS constellations: GPS, GLONASS, Galileo, BDS, and QZSS, plus SBAS satellites. If low power consumption is a key factor, then the module can be configured for a subset of GNSS constellations.

QZSS is a regional navigation satellite system that transmits signals compatible with the GPS L1 C/A, L1C, L2C and L5 signals for the Pacific region covering Japan and Australia. The module can detect and track QZSS L1 C/A and L5 signals concurrently with GPS signals, leading to better availability especially under challenging conditions, e.g., in urban canyons.

Table 4: GNSS Constellations and Frequency Bands

System	Signal
GPS	L1 C/A: 1575.42 MHz L5: 1176.45 MHz
GLONASS	L1: 1602 MHz + $K \times 562.5$ kHz K= (-7 to +6, integer)
Galileo	E1: 1575.42 MHz E5a: 1176.45 MHz
BDS	B1I: 1561.098 MHz B2a: 1176.45 MHz
QZSS	L1 C/A: 1575.42 MHz L5: 1176.45 MHz

1.6. Augmentation System

1.6.1. SBAS

The module supports SBAS signal reception. By augmenting primary GNSS constellations with additional satellite-broadcast messages, the system improves the accuracy and reliability of GNSS information by correcting signal measurement errors and providing information about signal accuracy, integrity, continuity, and availability. SBAS transmits signals for ranging or distance measurement, thus further improving availability. Supported SBAS systems: WAAS, EGNOS, MSAS and GAGAN.

1.7. AGNSS

The module supports AGNSS feature that significantly reduces the module's TTFF, especially under lower signal conditions. To implement AGNSS feature, the module should get the assistance data including the current time and rough position. For more information, see [document \[2\] AGNSS application note](#).

1.7.1. EASY

The module supports the EASY technology to improve TTFF by providing ancillary information, such as the ephemeris and almanac.

The EASY technology works as an embedded software to accelerate TTFF by predicting satellite navigation messages from the received ephemeris. After receiving the broadcast ephemeris for the first time, the GNSS engine automatically calculates and predicts orbit information for up to 3 subsequent

days, and saves the predicted information in the internal memory. The GNSS engine will use the information for positioning if there is not enough information from satellites, resulting in improved positioning and TTFF.

The EASY function reduces TTFF to 2 s in warm start. In this case, the backup domain should still be valid. To obtain enough broadcast ephemeris information from GNSS satellites, in strong-signal environments the GNSS module should keep tracking the information for at least 5 minutes after fixing the position.

The EASY function is enabled by default, and it can be disabled by **\$PAIR490**. For more information about command, see [document \[1\] protocol specification](#).

1.7.2. EPO

The module features a leading AGNSS technology called EPO, which assists the receiver to reduce the TTFF for up to 14 days. For more information about EPO, see [document \[2\] AGNSS application note](#).

1.8. Multi-Tone AIC

The module features a function called multi-tone active interference cancellation (AIC) to decrease the harmonic distortion of RF signals from Wi-Fi, Bluetooth, 2G, 3G, 4G and 5G networks.

Up to 12 AIC tones embedded in the module provide effective narrow-band interference and jamming elimination. Thus, the GNSS signal can be demodulated from the jammed signal, which can ensure better navigation quality.

The AIC function is enabled by default, and it can be disabled with **\$PAIR074**. For more information, see [document \[1\] protocol specification](#).

1.9. Geofencing

The module supports geofence areas, defined on the Earth's surface using a 2D model. Geofencing is active when at least one geofence area is defined. The current status can be found by polling the receiver. The receiver evaluates whether the current location of each region is within that region or not and signals its status via the GEOFENCE pin or **\$PQTMGEOFENCESTATUS**.

The evaluation is activated whenever one or more geofences are configured. For more information about geofencing configuration, see [document \[1\] protocol specification](#).

1.10. Firmware Upgrade

The module is delivered with preprogrammed firmware. Quectel may release firmware versions that contain bug fixes or performance optimizations. It is highly important to implement a firmware upgrade mechanism in your system. A firmware upgrade is the process of transferring a binary file image to the receiver and storing it in non-volatile flash. For more information, see [document \[3\] firmware upgrade guide](#).

2 Pin Assignment

The module matches industry convention and is equipped with 28 pins. Specifically, there are 18 LCC pins (Pins 1–18) on each side by which the module can be mounted onto legacy designs, and 10 LGA pins (Pins 19–28) under the top and bottom of the module which can provide access to additional advanced functionality.

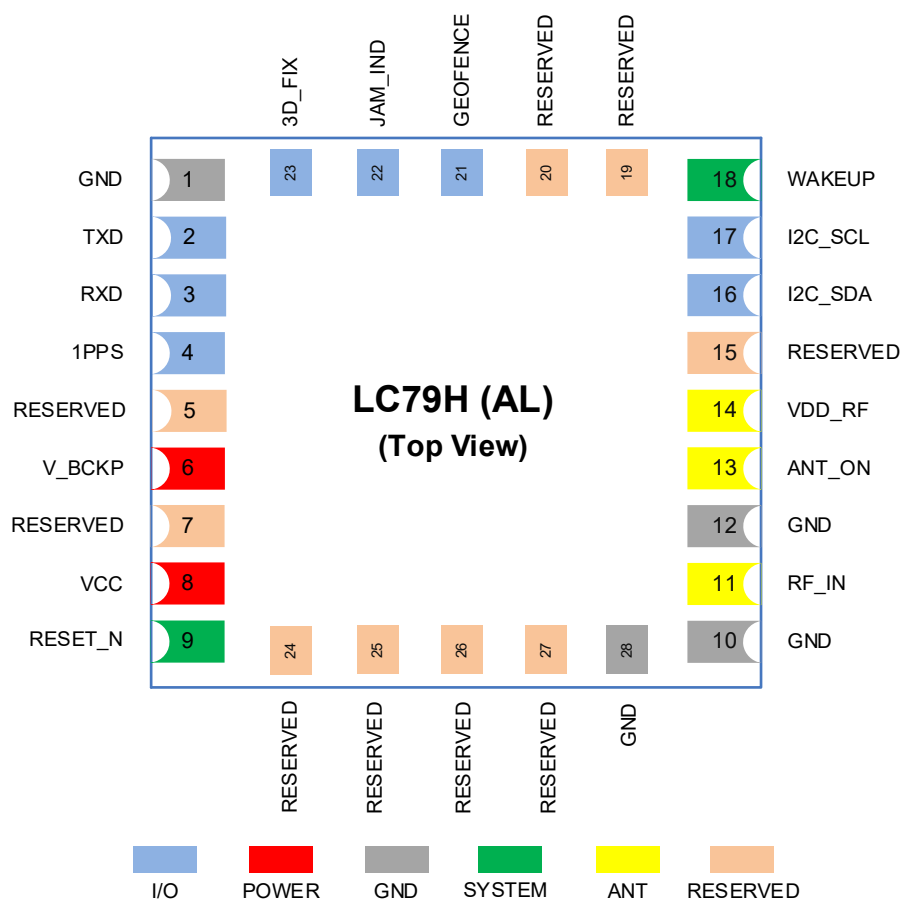


Figure 2: Pin Assignment

Table 5: I/O Parameter Definition

Type	Description
AI	Analog Input
DI	Digital Input
DO	Digital Output
DIO	Digital Input/Output
PI	Power Input
PO	Power Output

Table 6: Pin Description

Function	Name	No.	I/O	Description	DC Characteristics	Remarks
Power	VCC	8	PI	Main power supply	$V_{Imin} = 1.75\text{ V}$ $V_{Inom} = 1.8\text{ V}$ $V_{Imax} = 1.98\text{ V}$	Provides clean and steady voltage.
	V_BCKP	6	PI	Backup power supply for backup domain	$V_{Imin} = 1.62\text{ V}$ $V_{Inom} = 1.8\text{ V}$ $V_{Imax} = 1.98\text{ V}$	V_BCKP must be connected to power supply for startup, and it should always be powered if hot (warm) start is needed.
I/O	TXD	2	DO	Transmits data	$V_{OHmin} = 0.75 \times VCC$ $V_{OLmax} = 0.25 \times VCC$	The UART interface supports RTCM and standard NMEA messages, PAIR/PQTM message, binary data and firmware upgrade.
	RXD	3	DI	Receives data	$V_{ILmin} = -0.3\text{ V}$ $V_{ILmax} = 0.35 \times VCC$ $V_{IHmin} = 0.65 \times VCC$ $V_{IHmax} = 1.98\text{ V}$	
	I2C_SCL	17	DI	I2C serial clock	$V_{ILmin} = -0.3\text{ V}$ $V_{ILmax} = 0.35 \times VCC$ $V_{IHmin} = 0.65 \times VCC$ $V_{IHmax} = 1.98\text{ V}$ $V_{OLmax} = 0.25 \times VCC$ $V_{OHmin} = 0.75 \times VCC$	The I2C interface supports RTCM and standard NMEA messages, PAIR/PQTM message and binary data and firmware upgrade.
	I2C_SDA	16	DIO	I2C serial data		

Function	Name	No.	I/O	Description	DC Characteristics	Remarks
	GEOFENCE	21	DO	Indicates geofence status	$V_{OHmin} = 0.75 \times VCC$ $V_{OLmax} = 0.25 \times VCC$	Once the pin is enabled, the receiver continuously compares its current position to the preset geofence area. If unused, leave the pin N/C (not connected).
	JAM_IND	22	DO	Jamming indication		If unused, leave the pin N/C.
	3D_FIX	23	DO	3D fix indication		If unused, leave the pin N/C.
	1PPS	4	DO	One pulse per second		Synchronized on the rising edge. If unused, leave the pin N/C.
ANT	RF_IN	11	AI	GNSS antenna interface	-	50 Ω characteristic impedance.
	VDD_RF	14	PO	Supplies power for external RF components	$V_{onom} = VCC$	$VDD_RF = VCC$. The output current capacity depends on VCC. Typically used for powering an external active antenna or LNA. If unused, leave the pin N/C.
	ANT_ON	13	DO	Power control for external active antenna	$V_{OHmin} = 0.75 \times VCC$ $V_{OLmax} = 0.25 \times VCC$	If unused, leave the pin N/C.
System	RESET_N	9	DI	Resets the module	$V_{ILmin} = -0.3 \text{ V}$ $V_{ILmax} = 0.1 \text{ V}$ $V_{IHmin} = 1.62 \text{ V}$ $V_{IHmax} = 1.98 \text{ V}$	Active low.
	WAKEUP	18	DI	Wakes up the module from the Backup mode	$V_{IHmin} = 1.62 \text{ V}$ $V_{IHnom} = 1.8 \text{ V}$ $V_{IHmax} = 1.98 \text{ V}$	It is pulled down internally. Keep the pin low when the module is in the Continuous mode and the Backup mode. Drive the pin high for at least 10 ms to exit the Backup mode. If unused, leave the pin

Function	Name	No.	I/O	Description	DC Characteristics	Remarks
						N/C.
GND	GND	1, 10, 12, 28	-	Ground	-	Ensure a good GND connection to all GND pins of the module, preferably with a large ground plane.
RESERVED	RESERVED	5, 7, 15, 19, 20, 24–27	-	Reserved	-	These pins must be left floating and cannot be connected to power or GND.

NOTE

Leave RESERVED and unused pins N/C.

3 Power Management

The module features a power optimized architecture with built-in autonomous energy saving capabilities to minimize power consumption at any given time. The receiver can be used in two operating modes: Backup mode for optimum power consumption, and Continuous mode for optimum performance.

3.1. Power Unit

VCC is the supply voltage pin of the module. It supplies the PMU which in turn supplies the entire system. The load current of the VCC pin varies according to VCC voltage level, processor load and satellite acquisition. It is important to supply sufficient current and make sure the power supply is clean and stable.

The V_BCKP pin supplies the backup domain, which includes RTC and RAM. To achieve quick startup and improve TTFF, the backup domain power supply should be valid during the interval. The V_BCKP supplies RAM that contains all the necessary GNSS data and some of the user configuration variables.

VDD_RF is an output pin equal in voltage to the VCC input. In the Continuous mode, VDD_RF supplies for an external active antenna or the LNA. Only if VCC is cut off, VDD_RF is turned off.

The module's internal power supply block diagram is shown below:

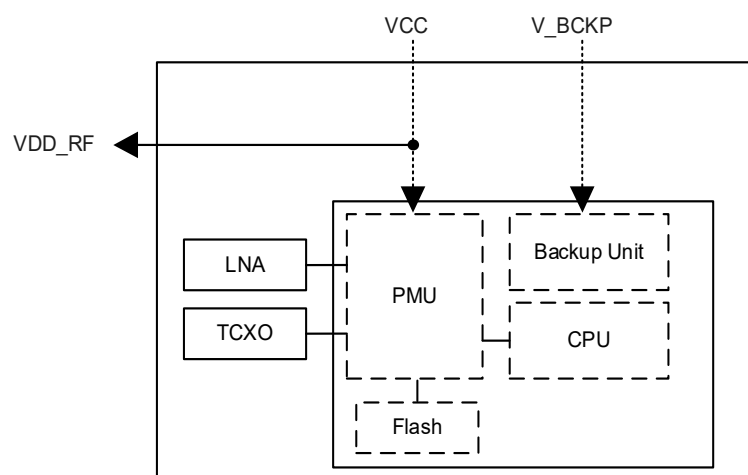


Figure 3: Internal Power Supply Block Diagram

3.2. Power Supply

3.2.1. VCC

The VCC is the supply voltage pin that supplies BB and RF.

Module power consumption may vary by several orders of magnitude, especially when a power saving mode is enabled. Therefore, it is important that the power supply is able to sustain peak power for a short time, ensuring that the load current does not exceed the rated value. When the module starts up or switches from the Backup mode to the Continuous mode, VCC must charge the internal capacitors in the core domain. In some cases, this can lead to a significant current drain.

For low-power applications using power saving mode, it is important that the LDO at the power supply or module input is able to provide sufficient current. An LDO with a high PSRR should be chosen for optimum performance. In addition, a TVS, and a combination of a 10 μF , a 100 nF and a 33 pF decoupling capacitor should be added near the VCC pin. The lowest value capacitor should be the closest to VCC pin.

It is not recommended to use a switching DC-DC converter.

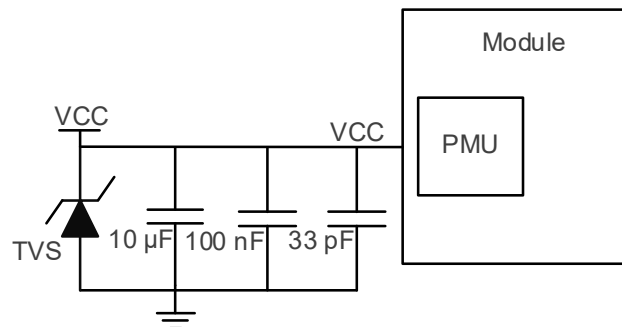


Figure 4: VCC Input Reference Circuit

NOTE

Ensure the module VCC is controlled by MCU to save power, or restart the module when it enters an abnormal state.

3.2.2. V_BCKP

The V_BCKP pin supplies the backup domain. Use of valid time and GNSS orbit data at startup allows GNSS hot (warm) start. V_BCKP must be connected to power supply for startup, and it should always be powered if hot (warm) start is needed.

If there is a constant power supply in your system, it can be used to provide a suitable voltage to power V_BCKP.

It is recommended to use a power supply of 1.8 V for V_BCKP and place a TVS and a combination of a 4.7 μ F, a 100 nF and a 33 pF capacitor near the V_BCKP pin. The reference circuit is illustrated below.

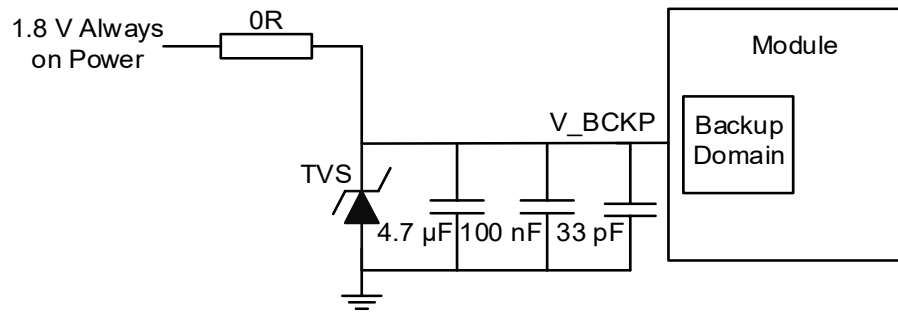


Figure 5: Backup Domain Reference Circuit

V_BCKP can also be powered by a 3.7 V lithium battery, as shown below.

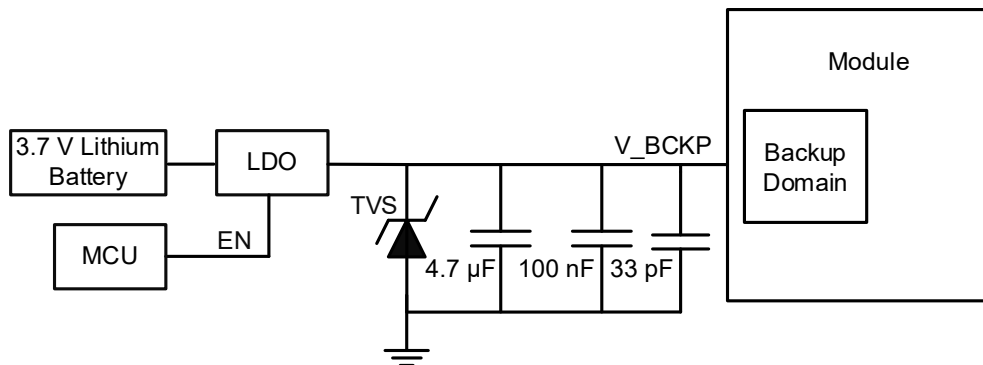


Figure 6: Reference Power Supply Circuit with 3.7 V Lithium Battery

NOTE

1. Using a non-rechargeable battery in Continuous mode is not recommended, as the maximum current consumption of V_BCKP exceeds 100 μ A and will deplete the battery.
2. If V_BCKP is below the minimum value of the recommended operating voltage, the module cannot work normally.
3. It is recommended to control the module V_BCKP via MCU to restart the module when the module enters an abnormal state.

3.3. Power Modes

3.3.1. Feature Comparison

The module features supported in different modes are listed in the table below.

Table 7: Feature Comparison in Different Power Modes

Features	Continuous	Backup
NMEA/RTCM from UART	●	-
1PPS	●	-
RF	●	-
Acquisition & Tracking	●	-
Power Consumption	High	Low
Position Accuracy	High	-

3.3.2. Continuous Mode

If VCC and V_BCKP are powered on, the module automatically enters the Continuous mode that comprises acquisition mode and tracking mode. In acquisition mode, the module starts to search satellites, and to determine visible satellites, coarse frequency, as well as the code phase of satellite signals. Once the acquisition is completed, the module automatically switches to tracking mode. In tracking mode, the module tracks satellites and demodulates the navigation data from specific satellites.

3.3.3. Backup Mode

For power-sensitive applications, the module receiver provides a Backup mode to reduce power consumption. Only backup domain is active in Backup mode and it keeps track of time.

- To enter Backup mode:
 1. Send **\$PAIR650** to shut down internal main powers in sequence.
 2. Cut off the power supply of the VCC pin and keep the V_BCKP pin powered.
- To exit Backup mode:
 1. Restore VCC power supply.
 2. Pull the WAKEUP pin high for at least 10 ms.

For more information about the command, see [document \[1\] protocol specification](#).

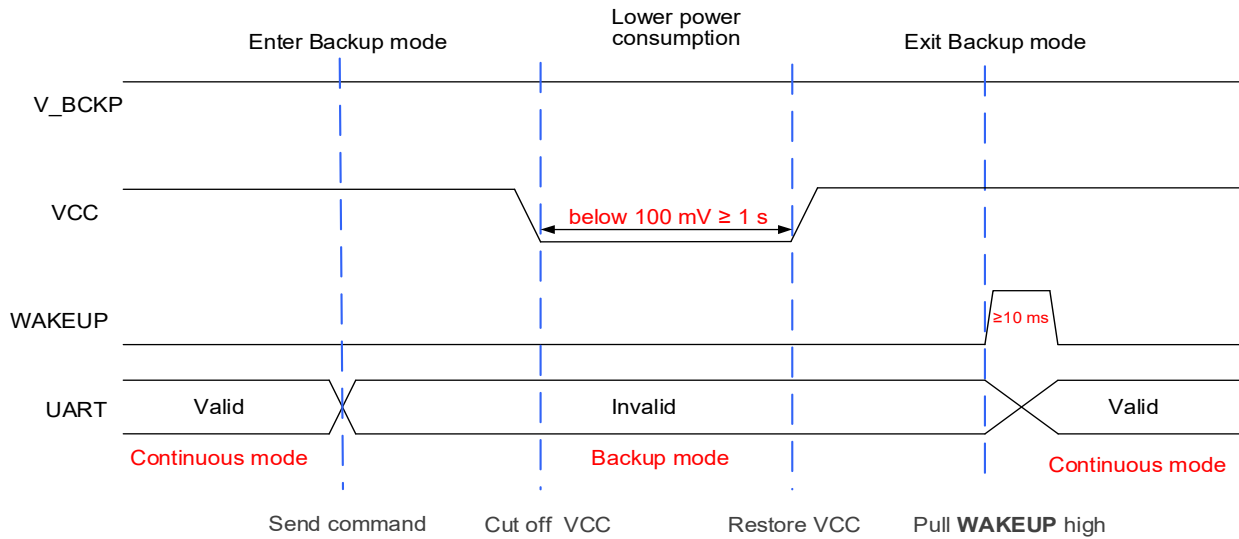


Figure 7: Enter/Exit Backup Mode Sequence

NOTE

1. **\$PAIR650** must be sent; to ensure a hot (warm) start at the module at the next startup, V_BCKP must be kept powered.
2. If the **\$PAIR650** is not sent or the VCC is not cut off, the module will not enter the Backup mode entirely, thus causing a higher current consumption than the specified value.
3. Keep the WAKEUP pin low in the Continuous mode and the Backup mode; drive the pin high to wake up the module from the Backup mode.
4. Ensure V_BCKP voltage stable without rush or drop when the VCC is switched on or off.

3.4. Power-up Sequence

Once the VCC and V_BCKP are powered up, the module starts up automatically and the voltage should rise rapidly in less than 50 ms.

To ensure the correct power-up sequence, the backup unit should start up no later than the PMU. Hence, the V_BCKP must be powered simultaneously with the VCC or before it.

Ensure that the VCC and V_BCKP have no rush or drop during the rising time, and then keep the voltage stable. The recommended ripple is < 50 mV.

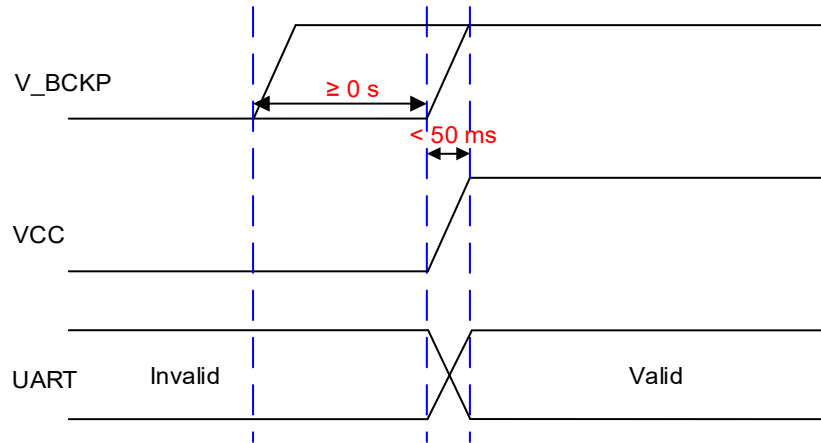


Figure 8: Power-up Sequence

3.5. Power-down Sequence

Once the VCC and V_BCKP are shut down, the module turns off automatically and voltage should drop quickly in less than 50 ms.

To avoid abnormal voltage conditions, if VCC and V_BCKP fall below the minimum specified value, the system must initiate a power-on restart by lowering VCC and V_BCKP to less than 100 mV for at least 1 s.

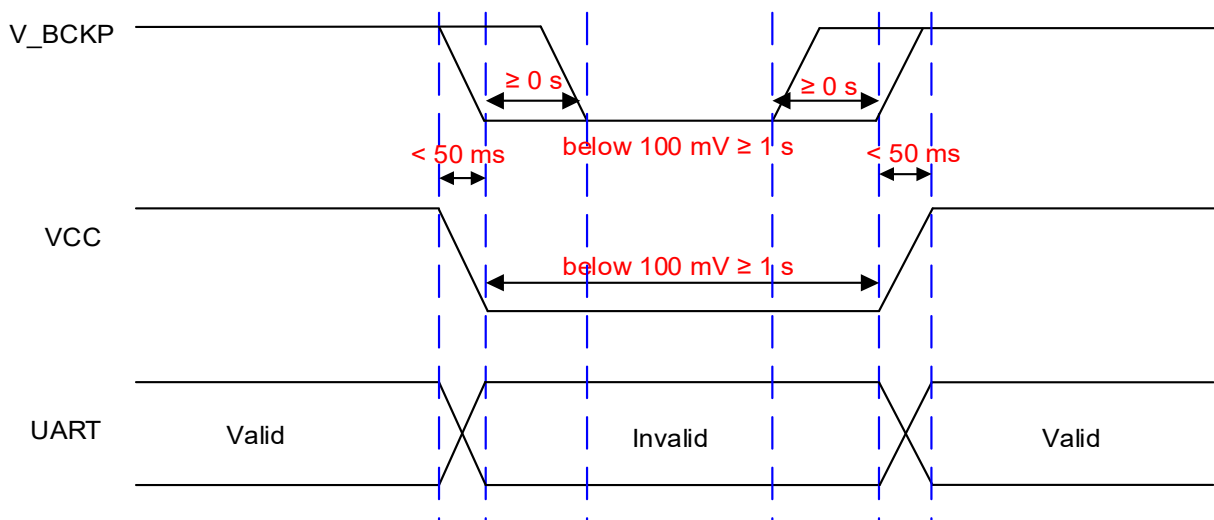


Figure 9: Power-down and Power-on Restart Sequence

4 Application Interfaces

4.1. I/O Pins

4.1.1. Communication Interfaces

The following interfaces can be used for data reception and transmission.

4.1.1.1. UART Interface

The module has one UART interface with the following features:

- Supports RTCM and standard NMEA messages, PAIR/PQTM message, binary data and firmware upgrade.
- Supported baud rates: 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600, and 3000000 bps.
- Hardware flow control and synchronous operation are not supported.

For more information, see [document \[1\] protocol specification](#).

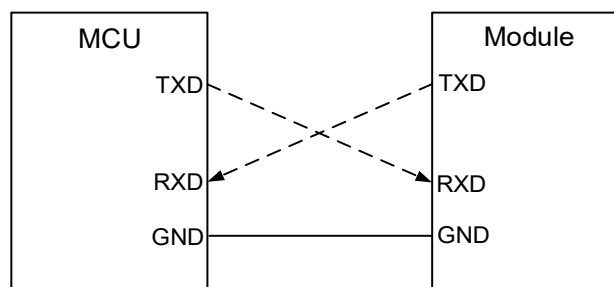


Figure 10: UART Interface Reference Design

A reference design is shown in the figure above. For more information, see [document \[4\] reference design](#).

NOTE

1. UART interface default settings may vary depending on software version. See the relevant software versions for details.
2. If the I/O voltage of MCU is not matched with that of module, a level-shifting circuit must be selected.

4.1.1.2. I2C Interface

The module has one I2C interface with the following features:

- Supports RTCM and standard NMEA messages, PAIR/PQTM message, binary data and firmware upgrade.
- Operates in slave mode.
- Supports 7-bit address.
- Supports the standard mode (100 kbps) and fast mode (400 kbps).

For more information, see [document \[1\] protocol specification](#).

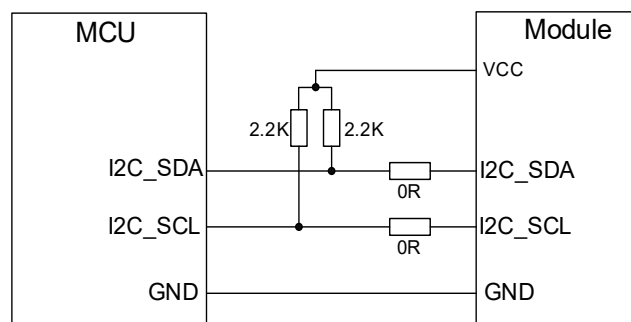


Figure 11: I2C Interface Reference Design

A reference design is shown in the figure above. For more information, see [document \[4\] reference design](#).

NOTE

If the I/O voltage of MCU is not matched with that of the module, a level-shifting circuit must be selected.

4.1.2. 3D_FIX

The 3D_FIX pin is at low level by default and assigned as a fix flag output. It outputs a high-level signal to indicate a successful 3D position fix.

4.1.3. GEOFENCE

The GEOFENCE pin indicates the current geofence status. Geofence configurations including geofence area(s) can be set using the related command. The receiver continuously compares its current position to the preset geofence areas(s) and the pin reflects whether the receiver is inside the active area(s) or not. It outputs a high logic level voltage to indicate that the receiver is inside the geofence area(s). For more information, see [document \[1\] protocol specification](#).

4.1.4. JAM_IND

In case of jamming that may interfere with the desired signal(s), the JAM_IND pin outputs a low-level signal; otherwise, it outputs a high-level signal.

4.1.5. 1PPS

The 1PPS output pin can be used for outputting time pulse signals as it generates one pulse per second periodic signal synchronized with a GNSS time grid with intervals. Maintaining high accuracy of 1PPS requires visible satellites in an open sky environment and powered VCC. See [Table 3: Product Performance](#) for details about pulse accuracy.

The 1PPS signal is output one pulse per second after successful positioning of the module. For more information, see [document \[1\] protocol specification](#).

4.2. System Pins

4.2.1. WAKEUP

The WAKEUP pin can be used to wake up the module from the Backup mode by being driven to a high voltage level for at least 10 ms. It is pulled down internally. Keep this pin at low voltage level in the Continuous mode and the Backup mode.

4.2.2. RESET_N

RESET_N is an input pin. The module can be reset by driving RESET_N pin low for at least 100 ms and then releasing it.

By default, the RESET_N pin is pulled up internally to 1.8 V with a 10 kΩ resistor, thus no external pull-up circuit is allowed for this pin.

An OC driver circuit as shown below is recommended to control the RESET_N pin.

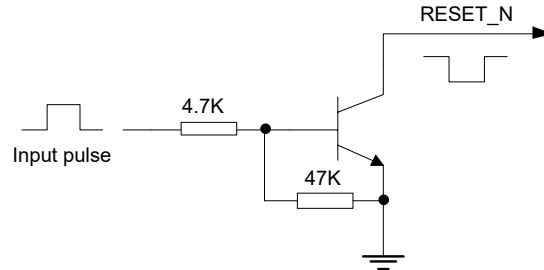


Figure 12: Reference OC Circuit for Module Reset

The following figure shows the reset sequence of the module.

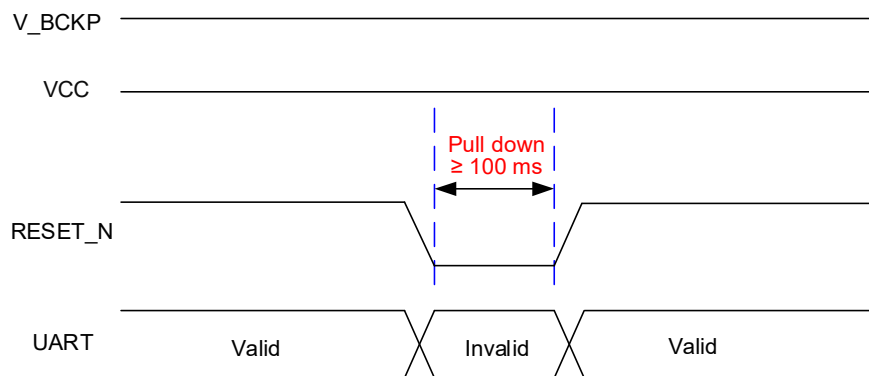


Figure 13: Reset Sequence

NOTE

RESET_N must be connected so that it can be used to reset the module if the module enters an abnormal state.

5 Design

This chapter explains the reference design of RF section and recommended footprint of the module.

5.1. Antenna Selection

5.1.1. Antenna Specifications

The module can be connected to a dedicated passive or active GNSS dual-band antenna to receive GNSS satellite signals. The recommended antenna specifications are given in the table below.

Table 8: Recommended Antenna Specifications

Antenna Type	Specifications
Passive Antenna	Frequency Range: 1164–1189 MHz & 1559–1606 MHz Polarization: RHCP VSWR: < 2 (Typ.) Passive Antenna Gain: > 0 dBi
Active Antenna	Frequency Range: 1164–1189 MHz & 1559–1606 MHz Polarization: RHCP VSWR: < 2 (Typ.) Passive Antenna Gain: > 0 dBi Active Antenna Noise Figure: < 1.5 dB Active Antenna Total Gain: < 17 dB

NOTE

1. For recommended antenna and design, see [document \[5\] GNSS antenna selection guidance](#) or contact Quectel Technical Support (support@Quectel.com).
2. The total antenna gain equals the internal LNA gain minus total insertion loss of cables and components inside the antenna.

5.1.2. Antenna Selection Guide

Both active and passive dual-band (L1 + L5) GNSS antennas can be used for the module. A passive antenna is recommended if the antenna can be placed close to the module, for instance, when the distance between the module and the antenna is less than 1 m. It is recommended to switch from a passive antenna to an active antenna once the loss is > 1 dB, since the insertion loss of RF cable can decrease the C/N₀ of GNSS signal. For more information about RF layout, see [document \[6\] RF layout application note](#).

C/N₀ is an important factor for GNSS receivers, and it is defined as the ratio of the received modulated carrier signal power to the received noise power in one Hz bandwidth. C/N₀ formula:

$$C/N_0 = \text{Power of GNSS signal} - \text{Thermal Noise} - \text{System NF(dB-Hz)}$$

The “Power of GNSS signal” is GNSS signal level. In practical environment, the signal level at the Earth’s surface is about -130 dBm. “Thermal Noise” is -174 dBm/Hz at 290 K. To improve C/N₀ of GNSS signal, an LNA could be added to reduce “System NF”.

“System NF”, formula:

$$NF = 10 \log F \text{ (dB)}$$

“F” is the noise factor of receiver system:

$$F = F1 + (F2 - 1)/G1 + (F3 - 1)/(G1 \cdot G2) + \dots$$

“F1” is the first stage noise factor, “G1” is the first stage gain, etc. This formula indicates that LNA with enough gain can compensate for the noise factor behind the LNA. In this case, “System NF” depends mainly on the noise figure of components and traces before first stage LNA plus noise figure of the LNA itself. This explains the need for using an active antenna, if the antenna connection cable is too long.

5.2. Antenna Reference Design

In a complex electromagnetic environment, a SAW filter circuit must be added to the antenna design to further reduce the impact of out-of-band signals on GNSS module. The SAW filter circuit has a stable suppression effect on all out-of-band signals. In the actual layout, the circuit should be placed close to RF_IN pin. The recommended SAW in the SAW filter circuit is B39162B2618P810 from RF360. SAW filter circuit can be selected according to the use case.

5.2.1. Active Antenna Reference Design

A typical reference design of an active antenna is illustrated in the following figure. In this case, the antenna is powered by VDD_RF. The active antenna with the SAW filter placed in front of the LNA in the internal framework must be used. This will minimize the impact of out-of-band signals on GNSS module performance in complex electromagnetic environments.

If VDD_RF pin is used to supply the active antenna, it is important to ensure that the operating voltage range of the antenna meets the voltage drop on the external power supply circuit, caused by the resistor (R2) and the inductor (L1). The minimum operating voltage of the selected active antenna must meet the circuit design characteristics.

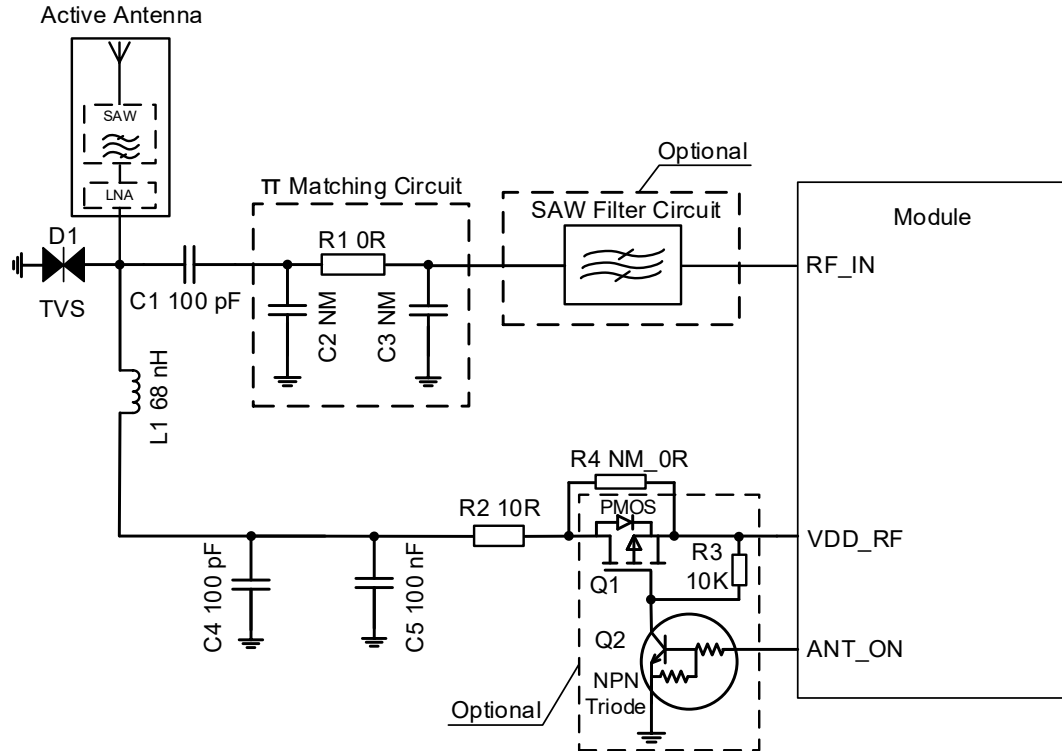


Figure 14: Active Antenna Reference Design

C1 is a DC-blocking capacitor used for blocking the DC current from VDD_RF. The C2, R1 and C3 components are reserved for matching antenna impedance. By default, C1 is 100 pF, R1 is 0 Ω, and C2 and C3 are not mounted. D1 is an electrostatic discharge (ESD) protection device to protect the RF signal input from the potential damage caused by ESD. The junction capacitance of D1 cannot be more than 0.6 pF and a transient voltage suppressor is recommended.

L1 inductor is used to prevent the RF signal from leaking into the VDD_RF and preventing noise propagation from the VDD_RF to the antenna. L1 inductor routes the bias voltage to the active antenna without losses. Place L1, C4 and C5 close to the antenna interface and route the proximal end of L1 pad on the RF trace. The recommended value of L1 should be at least 68 nH. R2 resistor is used to protect the module in case the active antenna is short-circuited to the ground plane.

The antenna is always powered when R4 is mounted. When it is not mounted, while Q1, Q2 and R3 are mounted, the antenna power supply can be controlled through the ANT_ON pin. When the pin outputs high level, the antenna is powered; otherwise, the antenna is not powered.

5.2.2. Passive Antenna Reference Design

The following figure is a typical reference design of a passive antenna.

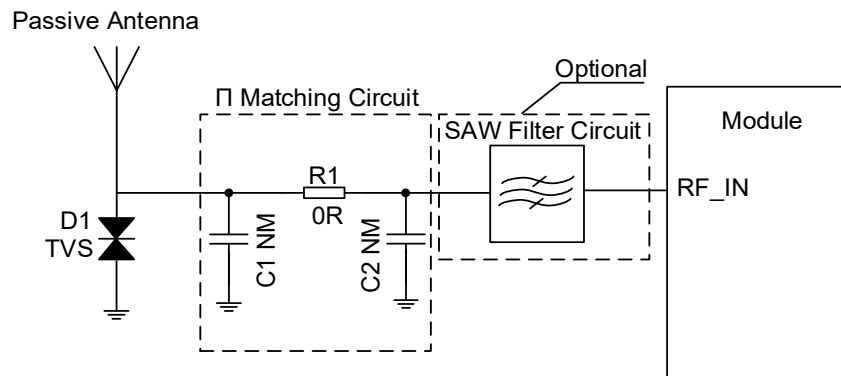


Figure 15: Passive Antenna Reference Design

C1, R1 and C2 are reserved for matching antenna impedance. By default, R1 is 0 Ω , while C1 and C2 are not mounted. D1 is an electrostatic discharge (ESD) protection device to protect RF route from the damage caused by ESD. The junction capacitance of D1 cannot be more than 0.6 pF and a transient voltage suppressor is recommended. The impedance of RF trace should be controlled to 50 Ω and trace length should be kept as short as possible.

5.3. Coexistence with Cellular Systems

Since GNSS signals are usually very weak, a GNSS receiver could be vulnerable to the environmental interference. According to 3GPP specifications, a cellular terminal should transmit a signal of up to 33 dBm at GSM bands, or of about 24 dBm at WCDMA and LTE bands, or of about 26 dBm at 5G bands. Therefore, coexistence with cellular systems must be optimized to avoid significant deterioration of the GNSS performance.

In a complex communication environment, interference may be caused by in-band and out-of-band signals as described in this chapter.

Suggestions for decreasing the impact of interference signals are also provided to ensure the interference immunity of a GNSS receiver.

5.3.1. In-band Interference

In-band interference refers to the signal whose frequency is within or near the operating frequency range of a GNSS signal. For example, GPS L1 is centered at 1575.42 MHz with a bandwidth of 2.046 MHz. As

shown in the figure below, the frequency of the interfering signal is within the GPS operation band, and the power of the interfering signal is higher than the power of the received GPS signal.

See the following figure for more details.

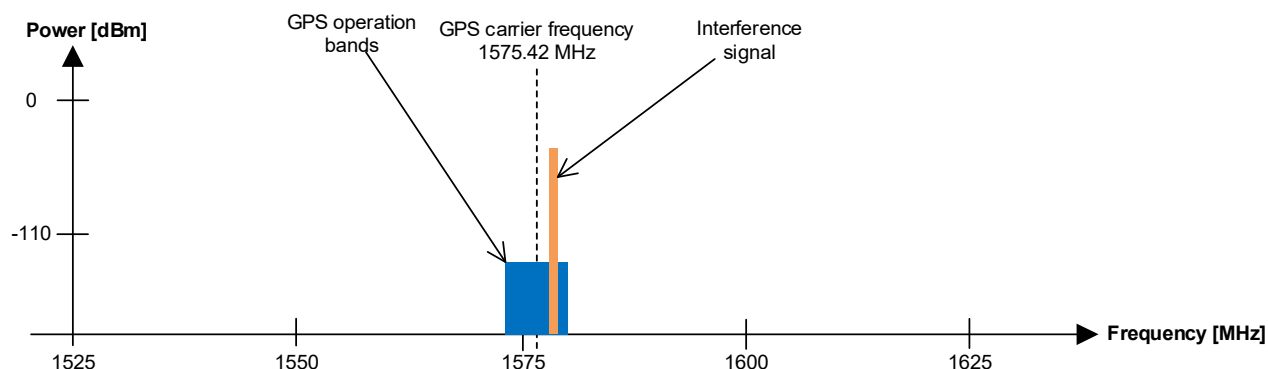


Figure 16: In-band Interference on GPS L1

In-band interferences are most commonly caused by:

- Harmonics, i.e., crystals, high-speed signal lines, MCUs, switch-mode power supply, or
- Intermodulation from different communication systems.

Common frequency combinations are presented in the table below. The table lists some probable in-band interferences generated by two kinds of out-of-band signal intermodulation, or the second harmonic of LTE Band 13.

Table 9: Intermodulation Distortion (IMD) Products

Source F1	Source F2	IM Calculation	IMD Products
GSM850/Band 5	Wi-Fi 2.4 GHz	$F2 (2412 \text{ MHz}) - F1 (837 \text{ MHz})$	IMD2 = 1575 MHz
Band 1	n78	$F2 (3500 \text{ MHz}) - F1 (1925 \text{ MHz})$	IMD2 = 1575 MHz
DCS1800/Band 3	PCS1900/Band 2	$2 \times F1 (1712.6 \text{ MHz}) - F2 (1850.2 \text{ MHz})$	IMD3 = 1575 MHz
PCS1900/Band 2	Wi-Fi 5 GHz	$F2 (5280 \text{ MHz}) - 2 \times F1 (1852 \text{ MHz})$	IMD3 = 1576 MHz
LTE Band 13	-	$2 \times F1 (786.9 \text{ MHz})$	IMD2 = 1573.8 MHz

5.3.2. Out-of-band Interference

Strong signals transmitted by other communication systems can cause GNSS receiver saturation, thus greatly deteriorating its performance, as illustrated in the following figure. In practical applications, common strong interference signals originate from wireless communication modules, such as GSM, 3G, LTE, 5G, Wi-Fi and Bluetooth.

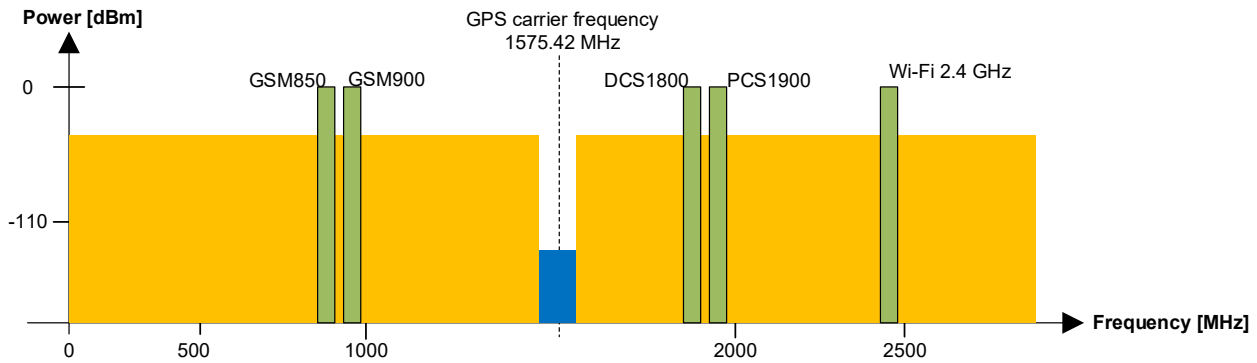


Figure 17: Out-of-band Interference on GPS L1

5.3.3. Ensuring Interference Immunity

There are several things you can do to decrease the impact of interference signals and thus ensure the interference immunity of a GNSS receiver:

- Keep the GNSS antenna away from interference sources;
- Add a band-pass filter in front of the GNSS module;
- Use shielding and multi-layer PCB and ensure adequate grounding;
- Optimize layout and component placement of the PCB and the whole device.

The following figure illustrates the interference source and its possible interference path. A complex communication system usually contains RF power amplifiers, MCUs, crystals, etc. These devices should be far away from a GNSS receiver, or a GNSS module. In particular, shielding should be used to prevent strong signal interference for power amplifiers. The cellular antenna should be placed away from a GNSS receiving antenna to ensure enough isolation. Usually, a good design should provide at least a 20 dB isolation between two antennas. Take DCS1800, for example, the maximum transmitted power of DCS1800 is around 30 dBm. After a 20 dB attenuation, the signal received by the GNSS antenna will be around 10 dBm, which is still too high for a GNSS module. With a GNSS band-pass filter with around 40 dB rejection in front of the GNSS module, the out-of-band signal will be attenuated to -30 dBm.

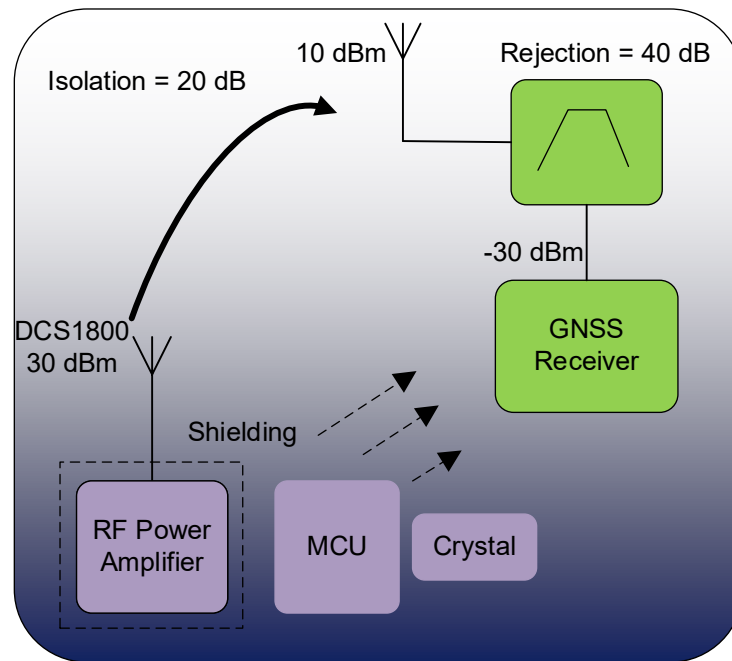


Figure 18: Interference Source and Its Path

5.4. Recommended Footprint

The figure below describes module footprint. These are recommendations, not specifications.

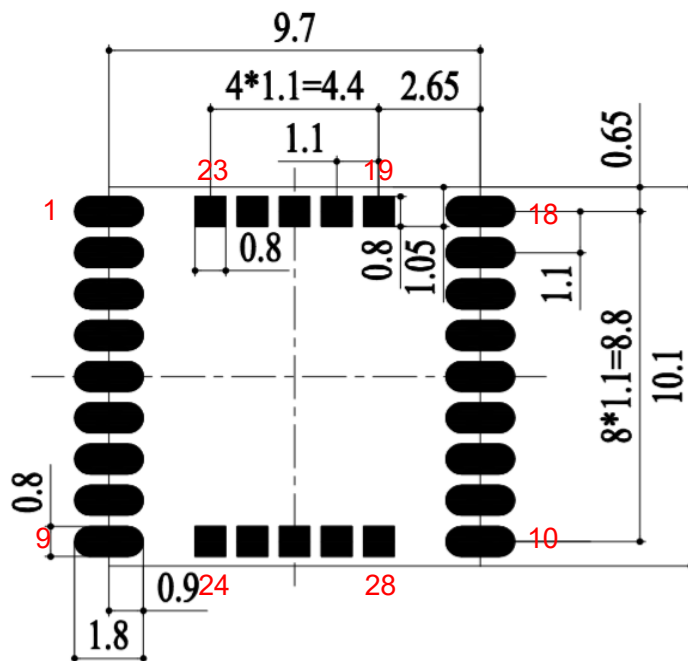


Figure 19: Recommended Footprint

NOTE

Maintain at least 3 mm keepout between the module and other components on the motherboard to improve soldering quality and maintenance convenience.

6 Electrical Specification

6.1. Absolute Maximum Ratings

Absolute maximum ratings for power supply and voltage on digital pins of the module are listed in table below.

Table 10: Absolute Maximum Ratings

Parameter	Description	Min.	Max.	Unit
VCC	Main Power Supply Voltage	-0.3	1.98	V
V_BCKP	Backup Power Supply Voltage	-0.3	1.98	V
V _{IN_IO}	Input Voltage at I/O Pins	-0.3	1.98	V
P _{RF_IN}	Input Power at RF_IN	-	0	dBm
T _{storage}	Storage Temperature	-40	90	°C

NOTE

Stressing the device beyond the “Absolute Maximum Ratings” may cause permanent damage. The product is not protected against over-voltage or reversed voltage. Therefore, it is necessary to use appropriate protection diodes to keep voltage spikes within the parameters given in the table above.

6.2. Recommended Operating Conditions

All specifications are at an ambient temperature of +25 °C. Extreme operating temperatures can significantly impact the specified values. Applications operating near the temperature limits should be tested to ensure specification validity.

Table 11: Recommended Operating Conditions

Parameter	Description	Min.	Typ.	Max.	Unit
VCC	Main Power Supply Voltage	1.75	1.8	1.98	V
V_BCKP	Backup Power Supply Voltage	1.62	1.8	1.98	V
IO_Domain	Digital I/O Pin Voltages Domain	-	VCC	-	V
V _{IL}	Digital I/O Pin Low-Level Input Voltage	-0.3	-	0.35 × VCC	V
V _{IH}	Digital I/O Pin High-Level Input Voltage	0.65 × VCC	-	1.98	V
V _{OL}	Digital I/O Pin Low-Level Output Voltage	-	-	0.25 × VCC	V
V _{OH}	Digital I/O Pin High-Level Output Voltage	0.75 × VCC	-	-	V
RESET_N	Low-Level Input Voltage	-0.3	-	0.1	V
	High-level Input Voltage	1.62	-	1.98	V
WAKEUP	High-Level Input Voltage	1.62	1.8	1.98	V
VDD_RF	VDD_RF Output Voltage	-	VCC	-	V
I _{VDD_RF}	VDD_RF Output Current	-	-	100	mA
T_operating	Operating Temperature	-40	25	+85	°C

NOTE

1. Operation beyond the “Operating Conditions” is not recommended and extended exposure beyond the “Operating Conditions” may affect device reliability.
2. The digital I/O pins in the table above refer to all digital pins in [Table 6: Pin Description](#) except RESET_N and WAKEUP.

6.3. Supply Current Requirement

The following table lists the supply current values of the total system that may be applied. Actual power requirements may vary depending on processor load, external circuits, firmware version, the number of tracked satellites, signal strength, startup type, test time and conditions.

Table 12: Supply Current

Parameter	Description	Condition	$I_{Typ.}^4$	I_{PEAK}^4
I_{VCC}^5	Current at VCC	Acquisition	33 mA	53 mA
		Tracking	33 mA	56 mA
$I_{V_BCKP}^6$	Current at V_BCKP	Continuous mode	74 μ A	146 μ A
		Backup mode	20 μ A	53 μ A

6.4. ESD Protection

Static electricity occurs naturally and it may damage the module. Therefore, applying proper ESD countermeasures and handling methods is imperative. For example, wear anti-static gloves during the development, production, assembly, and testing of the module; add ESD protective components to the ESD sensitive interfaces and points in the product design.

Measures to ensure protection against ESD damage when handling the module:

- When mounting the module onto a motherboard, make sure to connect the GND first, and then the RF_IN pin.
- When handling the RF_IN pin, do not come into contact with any charged capacitors or materials that may easily generate or store charges (such as patch antenna, coaxial cable, and soldering iron).
- When soldering the RF_IN pin, make sure to use an ESD safe soldering iron (tip).

⁴ Room temperature, measurements are taken with typical voltage.

⁵ Used to determine maximum current capability of power supply.

⁶ Used to determine required battery current capability.

7 Mechanical Dimensions

This chapter describes the mechanical dimensions of the module. All dimensions are in millimeters (mm). The dimensional tolerances are ± 0.20 mm, unless otherwise specified.

7.1. Top, Side and Bottom View Dimensions

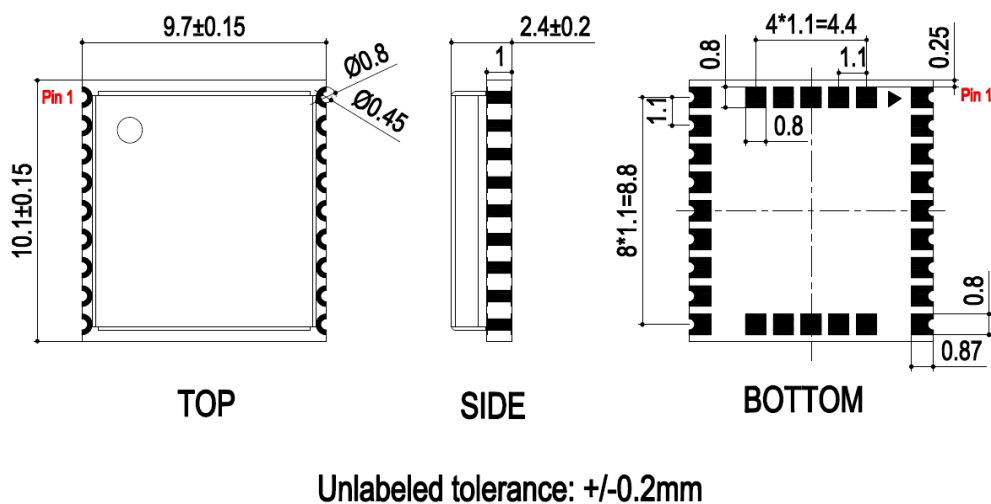


Figure 20: Top, Side and Bottom View Dimensions

NOTE

The package warpage level of the module conforms to the *JEITA ED-7306* standard.

7.2. Top and Bottom Views

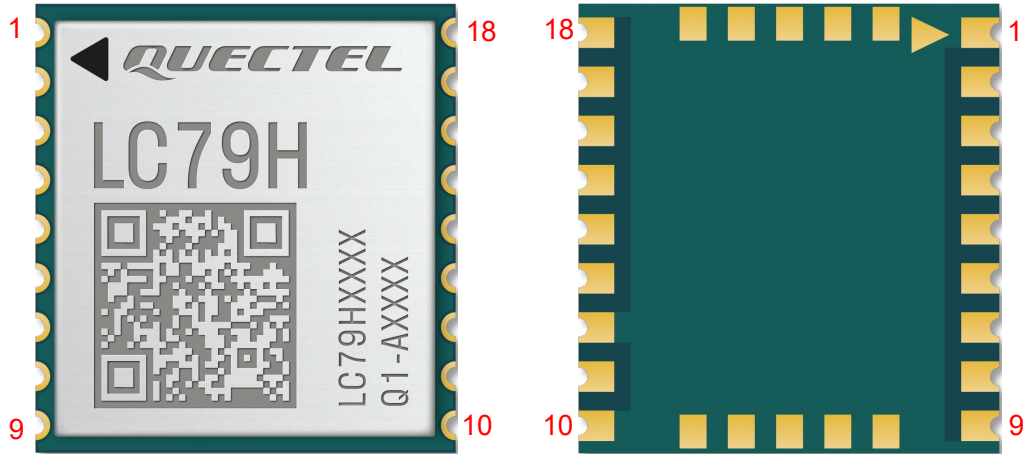


Figure 21: Top and Bottom Views

NOTE

The images above are for illustrative purposes only and may differ from the actual module. For authentic appearance and label, see the module received from Quectel.

8 Product Handling

8.1. Packaging

This chapter describes only the key parameters and process of packaging. All figures below are for reference only. The appearance and structure of packaging materials are subject to the actual delivery.

The module is packed in carrier tape packaging and the details are as follows.

8.1.1. Carrier Tape

Carrier tape dimensions are detailed below:

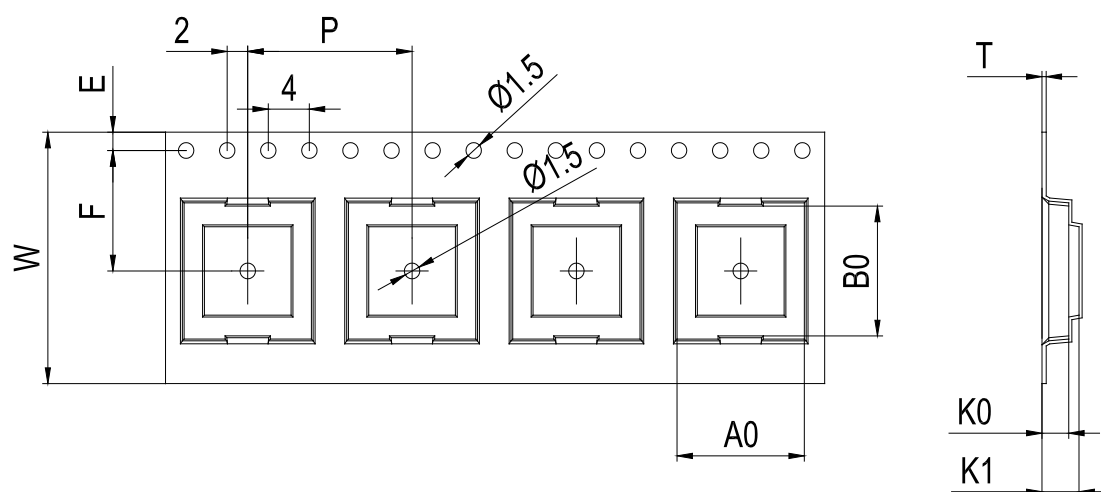


Figure 22: Carrier Tape Dimension Drawing

Table 13: Carrier Tape Dimension Table (Unit: mm)

W	P	T	A0	B0	K0	K1	F	E
24	16	0.3	10.1	10.5	2.8	3.3	11.5	1.75

8.1.2. Plastic Reel

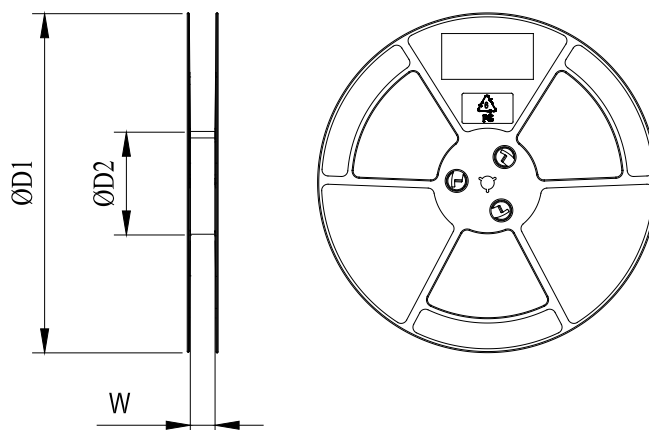


Figure 23: Plastic Reel Dimension Drawing

Table 14: Plastic Reel Dimension Table (Unit: mm)

ØD1	ØD2	W
330	100	24.5

8.1.3. Mounting Direction

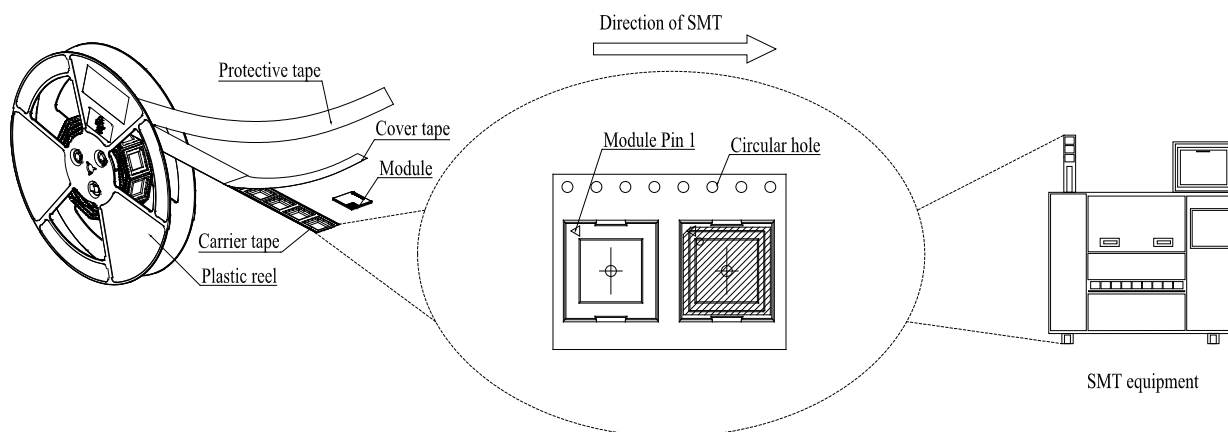
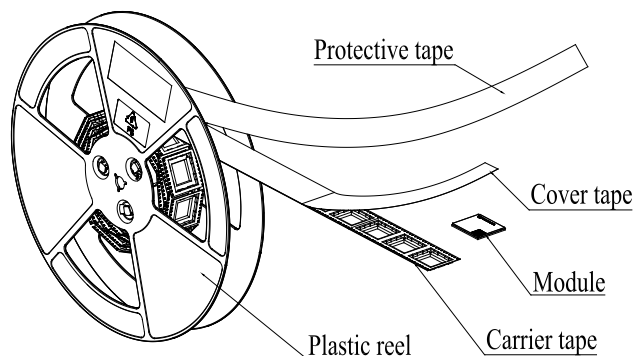


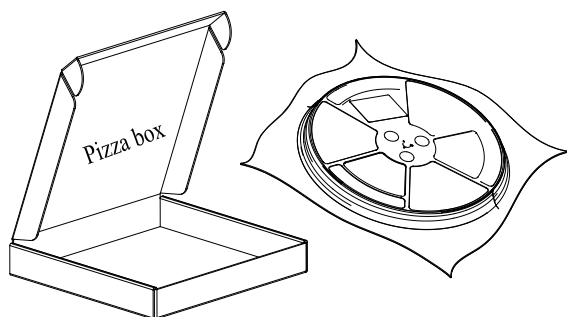
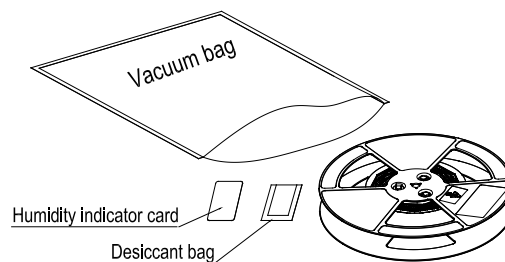
Figure 24: Mounting Direction

8.1.4. Packaging Process



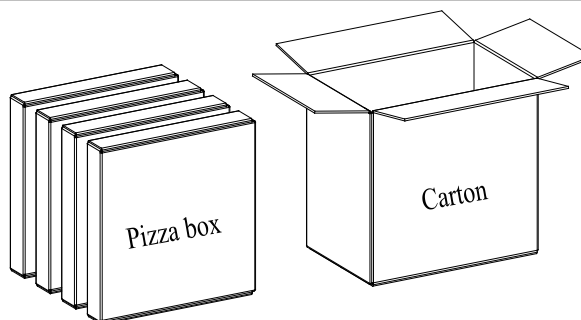
Place the module onto the carrier tape and use the cover tape to cover it; then wind the heat-sealed carrier tape on the plastic reel and use the protective tape for protection. 1 plastic reel can load 500 modules.

Place the packaged plastic reel, 1 humidity indicator card and 1 desiccant bag inside a vacuum bag, then vacuumize it.



Place the vacuum-packed plastic reel inside a pizza box.

Place 4 packaged pizza boxes inside 1 carton and seal it. 1 carton packs 2000 modules.



Pizza box size (mm): 363 × 343 × 41
Carton size (mm): 380 × 190 × 365

Figure 25: Packaging Process

8.2. Storage

The module is provided in the vacuum-sealed packaging. MSL of the module is rated as 3. The storage requirements are listed below.

1. Recommended Storage Condition: the temperature should be 23 ± 5 °C and the relative humidity should be 35–60 %.
2. Shelf life (in vacuum-sealed packaging): 12 months in Recommended Storage Condition.
3. Floor life: 168 hours ⁷ in a factory where the temperature is 23 ± 5 °C and relative humidity is below 60 %. After the vacuum-sealed packaging is removed, the module must be processed in reflow soldering or other high-temperature operations within 168 hours. Otherwise, the module should be stored in an environment where the relative humidity is less than 10 % (e.g., a dry cabinet).
4. The module should be pre-baked to avoid blistering, cracks and inner-layer separation in PCB under the following circumstances:
 - The module is not stored under Recommended Storage Condition;
 - Violation of the third requirement above;
 - Vacuum-sealed packaging is broken, or the packaging has been removed for over 24 hours;
 - Before module repairing.
5. If needed, the pre-baking should follow the requirements below:
 - The module should be baked for 8 hours at 120 ± 5 °C;
 - The module must be soldered to PCB within 24 hours after the baking, otherwise it should be put in a dry environment such as a dry cabinet.

NOTE

1. To avoid blistering, layer separation and other soldering issues, extended exposure of the module to the air is forbidden.
2. Take the module out of the package and put it on high-temperature-resistant fixtures before baking. If shorter baking time is desired, see *IPC/JEDEC J-STD-033* for the baking procedure.
3. Pay attention to ESD protection, such as wearing anti-static gloves, when touching the module.

⁷ This floor life is only applicable when the environment conforms to *IPC/JEDEC J-STD-033*. It is recommended to start the solder reflow process within 24 hours after the package is removed if the temperature and moisture do not conform to, or are not sure to conform to *IPC/JEDEC J-STD-033*. Do not unpack the modules in large quantities until they are ready for soldering.

8.3. Manufacturing and Soldering

Push the squeegee to apply solder paste on the surface of stencil, thus making the paste fill the stencil openings and then penetrate the PCB. Apply proper force on the squeegee to produce a clean stencil surface on a single pass. For more information about the stencil thickness for the module, see [document \[7\] module SMT application note](#).

The recommended peak reflow temperature should be 235–246 °C, with 246 °C as the absolute maximum reflow temperature. To avoid module damage caused by repeated heating, it is recommended to mount the module only after reflow soldering the other side of the PCB. The recommended reflow soldering thermal profile (lead-free reflow soldering) and related parameters are shown in the figure and table below.

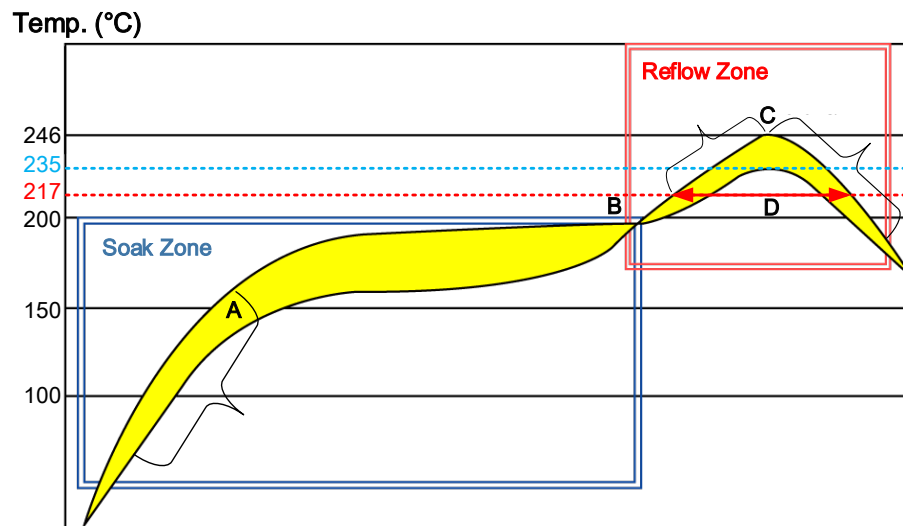


Figure 26: Recommended Reflow Soldering Thermal Profile

Table 15: Recommended Thermal Profile Parameters

Factor	Recommendation
Soak Zone	
Ramp-to-soak Slope	0–3 °C/s
Soak Time (between A and B: 150 °C and 200 °C)	70–120 s
Reflow Zone	
Ramp-up Slope	0–3 °C/s
Reflow Time (D: over 217 °C)	40–70 s
Max. Temperature	235–246 °C
Cooling Down Slope	-3–0 °C/s
Reflow Cycle	
Max. Reflow Cycle	1

NOTE

1. The above profile parameter requirements are for the measured temperature of the solder joints. Both the hottest and coldest spots of solder joints on the PCB should meet the above requirements.
2. During manufacturing and soldering, or any other processes that may require direct contact with the module, **NEVER** wipe the module shielding can with organic solvents, such as acetone, ethyl alcohol, isopropyl alcohol, and trichloroethylene. Otherwise, the shielding can may become rusty.
3. The module shielding can is made of cupronickel base material. The Neutral Salt Spray Test has shown that after 12 hours the laser-engraved label information on the shielding can is still clearly identifiable and the QR code is still readable, although white rust may be found.
4. If a conformal coating is necessary for the module, **DO NOT** use any coating material that may react with the PCB or shielding cover. Prevent the coating material from penetrating the module shield.
5. Avoid using ultrasonic technology for module cleaning since it can damage crystals inside the module.
6. Due to SMT process complexity, contact Quectel Technical Support in advance regarding any ambiguous situation, or any process (e.g., selective soldering, ultrasonic soldering) that is not addressed in [document \[7\] module SMT application note](#).

9 Labelling Information

The label of the Quectel GNSS modules contains important product information. The location of the product type number is shown in figure below.

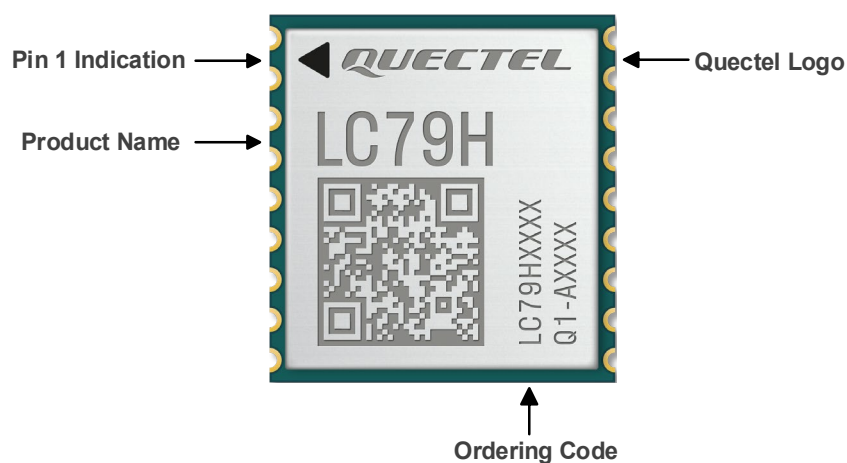


Figure 27: Labelling Information

The image above is for illustrative purposes only and may differ from the actual module. For authentic appearance and label, see the module received from Quectel.

10 Appendix References

Table 16: Related Documents

Document Name
[1] Quectel_LC29H_Series&LC79H(AL)_GNSS_Protocol_Specification
[2] Quectel_L89_R2.0&LC29H&LC79H_AGNSS_Application_Note
[3] Quectel_L89_R2.0&LC29H_Series&LC79H(AL)_Firmware_Upgrade_Guide
[4] Quectel_LC79H(AL)_Reference_Design
[5] Quectel_GNSS_Antenna_Selection_Guidance
[6] Quectel_RF_Layout_Application_Note
[7] Quectel_Module_SMT_Application_Note

Table 17: Terms and Abbreviations

Abbreviation	Description
AGNSS	Assisted GNSS (Global Navigation Satellite System)
AIC	Active Interference Cancellation
BDS	BeiDou Navigation Satellite System
bps	bit(s) per second
CEP	Circular Error Probable
C/N ₀	Carrier-to-noise Ratio
DCS1800	Digital Cellular System at 1800MHz
DR	Dead Reckoning
EASY	Embedded Assist System

Abbreviation	Description
EGNOS	European Geostationary Navigation Overlay Service
EPO	Extended Prediction Orbit
ESD	Electrostatic Discharge
GAGAN	GPS Aided Geo Augmented Navigation
Galileo	Galileo Satellite Navigation System (EU)
GLONASS	Global Navigation Satellite System (Russia)
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSM	Global System for Mobile Communications
I/O	Input/Output
I2C	Inter-Integrated Circuit
IC	Integrated Circuit
IMU	Inertial Measurement Unit
I _{PEAK}	Peak Current
NavIC	Indian Regional Navigation Satellite System
kbps	kilobits per second
LCC	Leadless Chip Carrier (package)
LDO	Low-dropout Regulator
LGA	Land Grid Array
LNA	Low-Noise Amplifier
LTE	Long-Term Evolution
MCU	Microcontroller Unit/Microprogrammed Control Unit
MSAS	Multi-functional Satellite Augmentation System (Japan)
MSL	Moisture Sensitivity Levels
NF	Noise Figure

Abbreviation	Description
NMEA	NMEA (National Marine Electronics Association) 0183 Interface Standard
OC	Open Connector
PCB	Printed Circuit Board
PI	Power Input
PMU	Power Management Unit
1PPS	1 Pulse Per Second
PSRR	Power Supply Rejection Ratio
QR (Code)	Quick Response (Code)
QZSS	Quasi-Zenith Satellite System
RAM	Random Access Memory
RF	Radio Frequency
RHCP	Right Hand Circular Polarization
RoHS	Restriction of Hazardous Substances
RTC	Real-Time Clock
RTK	Real-Time Kinematic
RXD	Receive Data (Pin)
3GPP	3rd Generation Partnership Project
SAW	Surface Acoustic Wave
SBAS	Satellite-Based Augmentation System
SMD	Surface Mount Device
SMT	Surface Mount Technology
SPI	Serial Peripheral Interface
TCXO	Temperature Compensated Crystal Oscillator
T_operating	Operating Temperature
TTFF	Time to First Fix

Abbreviation	Description
TVS	Transient Voltage Suppressor
TXD	Transmit Data (Pin)
UART	Universal Asynchronous Receiver/Transmitter
VCC	Supply Voltage
$V_{I\max}$	Maximum Input Voltage
$V_{I\min}$	Minimum Input Voltage
V_{Inom}	Normal Input Voltage
$V_{IH\max}$	High-level Maximum Input Voltage
$V_{IH\min}$	High-level Minimum Input Voltage
V_{IHnom}	High-level Normal Input Voltage
$V_{IL\max}$	Low-level Maximum Input Voltage
$V_{IL\min}$	Low-level Minimum Input Voltage
V_{Onom}	Normal Output Voltage
$V_{OL\max}$	Low-level Maximum Output Voltage
$V_{OH\min}$	High-level Minimum Output Voltage
VSWR	Voltage Standing Wave Ratio
WAAS	Wide Area Augmentation System
WCDMA	Wideband Code Division Multiple Access
XTAL	External Crystal Oscillator